Report On

VCET Lost & Found Portal

Submitted in partial fulfillment of the requirements of the Mini Project in Semester IV of Second Year Computer Science Engineering (Data Science)

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CERTIFICATE

This is to certify that the Mini Project entitled **"VCET Lost & Found Portal"** is a bonafide work of **Malati Mahadik (22), Niharika Mahajan (23), Mohammad Shuja Sayed (33), Swayam Lokare (20)** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of **"Bachelor of Engineering"** in Semester IV of Second Year **"Computer Science Engineering (Data Science)"**.

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Mini Project Approval

This Mini Project entitled **"VCET Lost & Found Portal"** by **Malati Mahadik (22), Niharika Mahajan (23), Mohammad Shuja Sayed (33), Swayam Lokare (20)** is approved for the degree of **Bachelor of Engineering** in Semester IV of Second Year **Computer Science Engineering (Data Science)**.

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Abstract

The VCET Lost & Found Portal is a web-based application developed to address the recurring challenge of lost item management within the Vidyavardhini's College of Engineering & Technology (VCET) campus. The existing approach of using physical notice boards, WhatsApp groups, and word-of-mouth is inefficient, lacks a centralized record, and often results in unclaimed items or failed reunions between owners and finders.

This project implements a structured, digital platform where students and staff can report lost or found items, browse listings, and contact relevant parties. The system is built using PHP and MySQL for backend processing and data storage, Bootstrap 5 for a responsive frontend, and Firebase Authentication for secure Google-based login restricted to @vcet.edu.in institutional accounts. It is hosted on InfinityFree (vcetlostfound.great-site.net) to ensure zero-cost public accessibility.

Key features include: image-supported item reporting, a Jaccard similarity-based automatic matching algorithm that cross-links lost and found items, owner-only delete permissions, a staff portal with password-based authentication for administrative oversight, and an item returned/resolved status toggle. The portal also supports a tap-to-fullscreen image lightbox for improved mobile usability.

The system successfully reduces dependence on informal communication channels, provides a searchable record of lost and found items, and improves the probability of item recovery through automated matching. Future enhancements may include push notifications, SMS alerts, and integration with the college's official student information system.

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List of Abbreviations

Abbreviation	Full Form
VCET	Vidyavardhini's College of Engineering & Technology
PHP	Hypertext Preprocessor
MySQL	My Structured Query Language
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DB	Database
UI	User Interface
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
NLP	Natural Language Processing
IDF	Inverse Document Frequency
TF	Term Frequency
AY	Academic Year
SRS	Software Requirements Specification
UML	Unified Modelling Language
ER	Entity-Relationship
GET	HTTP GET method
POST	HTTP POST method
URL	Uniform Resource Locator
SMTP	Simple Mail Transfer Protocol

Chapter 1

Introduction

1.1 Problem Statement

Campus environments such as colleges and universities witness a significant volume of lost and found incidents on a daily basis. Students and staff frequently misplace or forget personal belongings — ranging from identity cards and mobile phones to bags, stationery, and clothing — across classrooms, laboratories, libraries, canteens, and common areas. At VCET Vasai, the existing method for handling such situations relies entirely on informal channels: physical notice boards, WhatsApp group messages, and verbal communication among peers. This approach is unstructured, lacks persistence, and places a heavy burden on the finder to manually spread information.

The core problem is the absence of a centralized, searchable, and accessible platform where lost and found information can be recorded, viewed, and resolved. As a result, many items go unreturned simply because the right owner and finder never connect. The college administration also lacks a mechanism to oversee unresolved cases or maintain a record of recovered items.

1.2 Aim and Objectives

Aim:
To design, develop, and deploy a web-based Lost & Found portal for VCET that provides a centralized platform for reporting, browsing, and resolving lost and found item cases on campus, with automated item matching, secure institutional login, and staff oversight capabilities.

Objectives:

1. Provide a centralised, always-accessible digital platform for reporting lost and found items on the VCET campus.
2. Implement secure authentication using Firebase Google Login restricted to only institutional accounts, ensuring only verified campus members can access the system.
3. Enable image-supported item reporting with categorisation (category, location, date, description) to improve item identification accuracy.
4. Develop a Jaccard similarity-based automatic matching algorithm that cross-links reported lost items with reported found items, reducing manual search effort.
5. Implement owner-only deletion of item reports to prevent unauthorised removal of records.
6. Build a staff portal with password-based authentication allowing administrative staff to view all reports, manage unresolved cases, and toggle item status to Returned.

7. Deploy the portal as a publicly accessible, zero-cost hosted web application to ensure campus-wide availability.

1.3 Project Scope

The VCET Lost & Found Portal is scoped as a web-based application exclusively serving the VCET campus community. The portal is accessible to any individual holding an active @vcet.edu.in Google account. The current version fully supports the following:

- Reporting of lost and found items with optional image uploads, category selection, location, date, and description fields.
- Browsing of all active lost and found item listings by any logged-in user.
- Jaccard similarity-based automatic matching of lost items with found items, displayed on each item detail page.
- Owner-only delete functionality, preventing other users from removing records they did not create.
- Staff portal with password authentication, offering an administrative view of all records and the ability to mark items as Returned.
- Tap-to-fullscreen image lightbox for mobile-friendly viewing of uploaded item photographs.
- Cookie-based session persistence (30-day) with fallback email retrieval to resolve mobile browser localStorage clearing issues.

Out of scope for the current version are: real-time push notifications or email alerts, native mobile applications, integration with the VCET ERP or student information system, and multi-college or multi-campus expansion. These are identified as potential future extensions.

Chapter 2

Literature Survey

2.1 Survey of Existing Systems

A review of existing systems for lost and found management in educational and public settings reveals that the domain is largely underserved by dedicated, purpose-built solutions. Most institutions rely on either informal communication or generic ticketing tools that are not tailored to item recovery workflows.

General lost and found platforms such as Chargerback, iLost, and Found.com are designed for large-scale public environments such as airports, hotels, and transit systems. These systems are commercial, subscription-based, and carry overhead unsuitable for a small college campus. They also do not support institutional login constraints or campus-specific categorisation.

Several universities have developed internal portals using proprietary content management systems. However, most of these lack automated item matching, image upload support, or staff-specific administrative views. A study by Kushwaha et al. (2020) on campus management systems highlights that most digital solutions at Indian engineering colleges focus on attendance and results, leaving auxiliary services such as lost property management unaddressed.

Research in text similarity and NLP provides the theoretical foundation for automated item matching. The Jaccard similarity coefficient, introduced by Paul Jaccard (1901) in the context of set theory, measures the overlap between two sets and has since been widely applied in information retrieval, plagiarism detection, and recommendation systems. Its use in this portal, applied to item description tokens, provides a lightweight, interpretable matching mechanism without requiring external libraries or machine learning models.

Table 2.1: Comparison of Existing Lost & Found Systems

System	Image Upload	Auto Matching	Auth Control	Campus-Specific
Chargerback	Yes	No	No	No (Commercial)
iLost	Yes	No	No	No (Commercial)
WhatsApp Groups	Yes	No	None	Informal

Physical Notice Board	No	No	None	Yes (Manual)
Generic CMS Portals	Limited	No	Basic	Partial
VCET L&F Portal (Ours)	Yes	Yes (Jaccard)	Institutional Google	Yes

2.2 Limitations of Existing Systems and Research Gap

The key limitations identified in the surveyed systems are as follows. First, no existing free-to-use solution provides campus-restricted access tied to institutional email domains, which is essential for trust and accountability in a college environment. Second, commercial platforms are cost-prohibitive for individual institutions with no IT procurement budget. Third, WhatsApp and notice board approaches lack persistence — items reported informally disappear from view as newer messages arrive and boards fill up. Fourth, no reviewed system implements automated text-based matching between lost and found records, requiring users to manually search through all listings. Fifth, administrative oversight features (staff portal, status toggling) are absent from informal systems and optional or enterprise-only in commercial platforms.

The research gap is the absence of a free, self-hosted, institutionally authenticated portal that: (a) provides persistent item records with image support, (b) automates similarity-based item matching, (c) enforces institutional access control, and (d) offers a staff-level administrative view all within a single zero-cost deployable application.

2.3 Mini Project Contribution

The VCET Lost & Found Portal makes the following distinct contributions:

8. Institutional Authentication Gate: Firebase Google login restricted to @vcet.edu.in accounts ensures that only verified VCET community members can submit or view reports, preventing spam and misuse.
9. Jaccard Similarity-Based Automatic Matching: Item descriptions are tokenised and compared using the Jaccard coefficient to surface potentially related lost and found reports, reducing manual search effort for users.
10. Owner-Controlled Data: Each item report is tied to the submitter's authenticated email; delete operations are restricted to the owner, preserving data integrity.
11. Staff Portal with Resolution Tracking: An administrative interface allows designated staff to view all records across all users and mark cases as Returned, providing institutional oversight and closure tracking.
12. Zero-Cost Deployment on InfinityFree: The portal is hosted on InfinityFree (vcetlostfound.great-site.net) at no cost, demonstrating that useful campus infrastructure can be built and deployed without institutional IT budget.

13. Mobile-Optimised UX: Bootstrap 5 responsive design, 30-day cookie-based session persistence, and a tap-to-fullscreen image lightbox ensure usability on mobile devices, which are the primary access medium for students.

Chapter 3

Proposed System

3.1 Introduction

The VCET Lost & Found Portal is built using a PHP and MySQL backend with a Bootstrap 5 responsive frontend. Authentication is handled by Firebase Authentication (Google OAuth 2.0) with domain restriction to @vcet.edu.in accounts. The application is hosted on InfinityFree, a free PHP and MySQL web hosting provider, accessible at vcetlostfound.great-site.net.

The backend consists of PHP scripts for form processing (report lost, report found), database CRUD operations, session management, and staff portal functions. MySQL stores all item records in two primary tables: lost_items and found_items. Firebase JavaScript SDK handles the client-side Google login flow, and the authenticated user's email is passed to PHP via GET parameters and stored in long-lived cookies for session continuity across page loads.

The matching algorithm operates server-side in PHP, computing Jaccard similarity scores between a viewed item's description tokens and all records in the opposite table (lost vs found), returning the top matches above a configurable threshold.

3.2 System Architecture and Block Diagram

The system architecture is organised into three layers: Client Layer, Server Layer, and Data Layer.

Client Layer: The user's web browser loads the Bootstrap 5 frontend. Firebase JavaScript SDK executes the Google login flow client-side, obtains the authenticated user email, and passes it to the PHP backend via GET parameters and cookies. All UI interactions (form submission, page navigation, image tap-to-fullscreen) are handled in the browser.

Server Layer: InfinityFree's PHP 7.4 server processes all HTTP requests. Individual PHP scripts handle: (a) login session initialisation, (b) report lost item POST with image upload, (c) report found item POST with image upload, (d) browse/search listings, (e) item detail view with Jaccard matching, (f) delete operations with ownership check, (g) staff portal authentication and management actions.

Data Layer: A MySQL database stores two primary tables — lost_items and found_items — each with fields for item ID, user email, category, title, description, location, date, image path, status (active/returned), and timestamp. File uploads are stored in a designated uploads/ directory on the server.

[Figure 3.1 — System Architecture Block Diagram to be inserted here]

[Figure 3.2 — ER Diagram to be inserted here]

3.3 Algorithm and Process Design

3.3.1 User Authentication Flow

On page load, the Firebase JavaScript SDK checks for an active Google session. If no session exists, the user is prompted to sign in with their Google account. Firebase enforces the @vcet.edu.in domain restriction at the OAuth level. Upon successful login, the authenticated email is extracted and stored in a 30-day browser cookie. On subsequent page loads, PHP reads this cookie via a getEmail() function to identify the current user without requiring a repeated login.

3.3.2 Item Reporting Flow

When a user submits the Report Lost or Report Found form, the data (title, category, location, date, description, optional image) is sent via HTTP POST to the corresponding PHP script. The user email is passed as a GET parameter appended to the form action URL to work around InfinityFree's restriction on accessing hidden POST fields from PHP. The PHP script validates inputs, moves the uploaded image to the uploads/ directory, and inserts a new record into the respective MySQL table.

3.3.3 Jaccard Similarity Matching Algorithm

When a user views a specific lost item, the system retrieves all active records from the found_items table (and vice versa for found items). For each candidate record, the Jaccard similarity between the viewed item's description and the candidate's description is computed as follows:

$$J(A, B) = |A \cap B| / |A \cup B|$$

Where A and B are the sets of lowercase tokens (words) extracted from each description after removing punctuation. A similarity score above a threshold of 0.15 qualifies a candidate as a match. The top matches are displayed on the item detail page as Possible Matches, with the similarity score shown as a percentage.

[Figure 3.6 — Jaccard Matching Flowchart to be inserted here]

3.4 Details of Hardware and Software

Hardware / Software Component	Specification / Version
Development Machine	Any PC/Laptop with internet access
Operating System	Windows 10/11 or Linux
Web Server	InfinityFree (Apache, PHP 7.4)
Database	MySQL 5.7 (InfinityFree hosted)
Backend Language	PHP 7.4
Frontend Framework	Bootstrap 5.3
Authentication	Firebase Authentication (Google OAuth 2.0)
Firebase SDK	Firebase JS SDK 9.x (compat mode)
Version Control	Git / GitHub
Browser (Client)	Google Chrome, Mozilla Firefox, Safari (mobile)
IDE / Editor	Visual Studio Code
Image Hosting	Server-side uploads/ directory on InfinityFree

3.5 Experiment and Results for Validation and Verification

The portal was tested by team members and a group of fellow students across both desktop (Chrome, Firefox) and mobile (Android Chrome, iOS Safari) browsers. The following test scenarios were executed:

- Authentication: Login with @vcet.edu.in accounts succeeded; login attempts with non-VCET Google accounts were correctly blocked.
- Report Lost / Found: Forms with and without image uploads were submitted successfully. Validation messages appeared correctly for missing required fields.
- Browse Listings: Lost and found items appeared correctly in tabular listings. Pagination and filtering by category functioned as expected.
- Item Matching: Test items with overlapping keywords in descriptions (e.g., 'blue bag class 3rd floor' vs 'found blue bag near third floor') returned match scores of 0.25-0.40, surfacing as Possible Matches.
- Owner Delete: The delete button appeared only for the item's owner. Attempting to delete another user's item via direct URL manipulation was rejected.
- Staff Portal: Staff login with the correct password granted access to all records and the Returned toggle. Incorrect passwords were rejected.
- Mobile Session Persistence: After clearing mobile browser cache, the getEmail() fallback function correctly retrieved the user's email from the 30-day cookie, maintaining session continuity.

[Figure 3.7–3.13 — Screenshots to be inserted here]

3.6 Analysis

The portal successfully addresses the primary objectives. Authentication restricts access to the VCET community, ensuring a trusted user base. The Jaccard matching algorithm, while simple, demonstrates effective results for short descriptive texts typical of lost and found reports — keyword overlap between descriptions of the same physical item is generally sufficient to produce a non-zero similarity score.

A notable technical challenge encountered and resolved during development was InfinityFree's restriction on PHP reading hidden POST form fields. This was worked around by passing the user email as a GET parameter in the form action URL. Another challenge was mobile browser localStorage clearing, which disrupted session continuity; this was resolved by switching to 30-day cookies with a getEmail() fallback and a setTimeout delay on page load.

The portal's zero-cost deployment on InfinityFree confirms the feasibility of providing useful campus infrastructure without institutional IT budgets. The hosted URL (vcetlostfound.great-site.net) is publicly accessible and has been validated across multiple device types.

3.7 Conclusion and Future Work

The VCET Lost & Found Portal achieves its core goal of providing a centralised, authenticated, and functional platform for managing lost and found items on the VCET campus. It replaces informal, unstructured communication channels with a persistent, searchable digital system that benefits both students and administrative staff.

The project demonstrates the practical application of web technologies (PHP, MySQL, Bootstrap 5, Firebase Authentication) in solving a real campus problem. The Jaccard similarity algorithm provides a lightweight, library-free matching mechanism that is appropriate for the scale and nature of item descriptions submitted by campus users.

Future Work: Planned enhancements include:

14. Email/push notifications to alert users when a potential match is found for their reported item.
15. Integration with the VCET student information system to auto-populate user details on report forms.
16. A more advanced matching algorithm using TF-IDF weighted similarity or word embedding cosine similarity for improved match quality.
17. A dedicated mobile app (Android/iOS) for faster access and camera-based image upload.
18. An admin analytics dashboard showing resolution rates, category breakdowns, and average time to resolution.

19. Multi-language support (Marathi/Hindi) to improve accessibility for non-English-proficient users.

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4. InfinityFree Hosting Documentation, https://support.infinityfree.com/

5. Firebase Authentication Docs, https://firebase.google.com/docs/auth/web/google-signin

Annexure

A. Published Paper / Camera Ready Paper / Proof of Concept

The VCET Lost & Found Portal was presented at the VNPS 2026 — Oscillations National Project Showcase (Track 6, Multidisciplinary). A research paper abstract and a 1920x1080 poster were submitted and accepted for display. The portal hosted live at vcetlostfound.great-site.net served as the proof of concept during the showcase.

B. Source Code Repository

The complete source code for the VCET Lost & Found Portal is maintained on GitHub. The repository includes all PHP scripts, frontend HTML/CSS/JS files, database schema SQL, and documentation.

Repository URL: [To be updated with GitHub link]

C. Database Schema

The MySQL database schema consists of the following tables:

lost_items (id INT AUTO_INCREMENT PRIMARY KEY, user_email VARCHAR(255), category VARCHAR(100), title VARCHAR(255), description TEXT, location VARCHAR(255), lost_date DATE, image_path VARCHAR(500), status ENUM('active','returned') DEFAULT 'active', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP)

found_items (id INT AUTO_INCREMENT PRIMARY KEY, user_email VARCHAR(255), category VARCHAR(100), title VARCHAR(255), description TEXT, location VARCHAR(255), found_date DATE, image_path VARCHAR(500), status ENUM('active','returned') DEFAULT 'active', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP)